

MEMO — Excelta @ DigiKey Scraper & Analysis

Date: 2026-06-30 · **Status:** Built, tested, live-verified

What it does

Pulls every Excelta product from DigiKey daily → dated CSV/XLSX → MBB-style analysis → (once history exists) trend/demand layer.

Why API, not scraping

DigiKey storefront is Cloudflare-protected (curl → 403). Built on **DigiKey Product Information API v4** instead — sanctioned, reliable for unattended daily runs. Resolved hard limits by testing: **Offset+Limit ≤ 300** per query, empty-keyword requires a category filter, manufacturer = "Excelta Corporation" (id 2827). Categories >300 (tweezers 556, Tools 1296) are recursively subdivided by parametric filter.

Latest run

1,330 unique parts, ~46s, 22 categories. CSV + formatted XLSX (Products + Summary).

Key findings (T=0 baseline)

- **Value Pareto-extreme:** 157 SKUs (24%) = 80% of \$501,745 visible inventory value.
- **Availability is the leak:** 51% stockout. Worst: screw/nut drivers 96%, wire cutters 76% (also #2 value line → priority).
- **Pricing wide, under-laddered:** median \$54.84, max \$2,427; knives category spread 788x; 150 quote-only (\$0) SKUs.

Deliverables

Component	Output
main.py	excelta_<date>.csv + .xlsx (scrape)
analyze.py	excelta_analysis_<date>.xlsx (6 sheets + Pareto/stockout charts) + ANALYSIS_<date>.md
timeseries.py	TRENDS_<date>.md + excelta_changes_<date>.csv (gated: needs ≥2 snapshots)
tests	15 offline tests pass

Discipline ("no mistakes")

Inventory value = list×stock **proxy** (not COGS). \$0 excluded from price stats; qty=0 a real state, never imputed. **No trend/demand claims on one snapshot** — WS6 dormant until day 2.

To operationalize

1. Cron `main.py` daily (line in README).
2. Run `analyze.py` for the deck; `timeseries.py` auto-fires once 2 snapshots exist.

Open items

- App may be Sandbox-tier — confirm production approval for full live data.
- **Rotate the API secret** (it appeared in build chat).
- Demand proxy can't separate sale from stock adjustment — averages out over time; don't read single-day spikes as sales.
- Competitive benchmarking needs peer-manufacturer snapshots (future extension).